

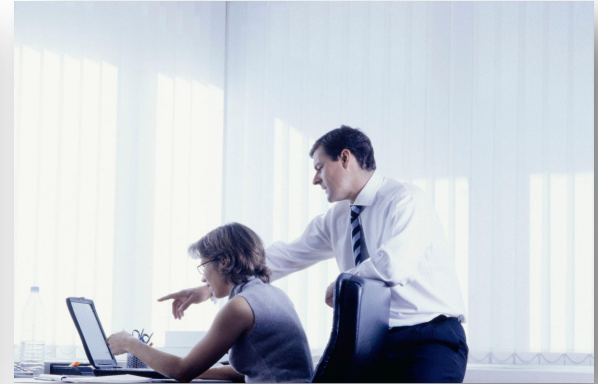
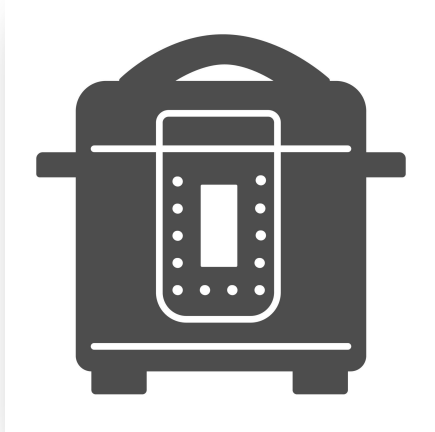
Week 5: Introduction to GenAI

Course: BUSM2725 AI and Analytics for Business Applications

Outline

- Overview of NLP, language models, and LLMs
- Activities: What LLMs (ChatGPT, Gemini) can do?
- Activity: How to control the responses of LLMs (temperature, structured outputs, prompt/context engineering)
- Demo: Introducing SPARK and GenAI Concepts in Action
- Assessment 2: Develop a Proposal for an AI Analytics Agent

What is AI to you?



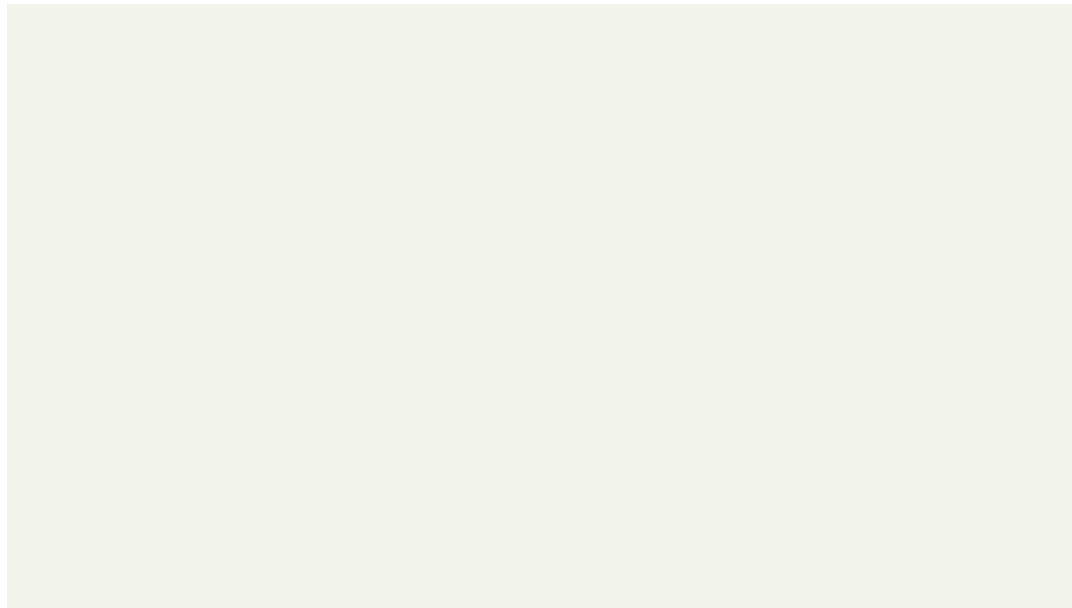
What LLMs (ChatGPT, Gemini) can do?

What AI can understand?	What AI can generate?
Texts	Texts
Images	Images
Documents (Texts + Images)	Videos
Videos/Youtube	Audio
Websites	Combinations of above
Audio/Live Streams	

Check out this example



Source: *Business
Council of Australia*



Source: *Generated with Google Vids*

What kind of content is generated by AI in this video?

How ChatGPT is developed?

Step 1

Collect demonstration data and train a supervised policy.

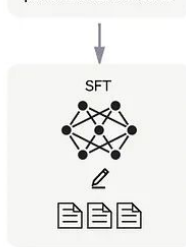
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3.5 with supervised learning.



Step 2

Collect comparison data and train a reward model.

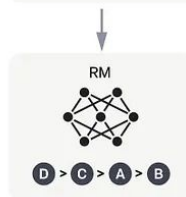
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



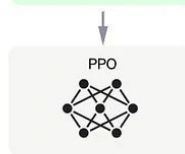
Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

A new prompt is sampled from the dataset.



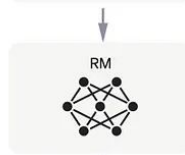
The PPO model is initialized from the supervised policy.



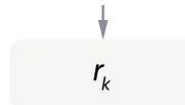
The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



Large language models

Generative AI applications are powered by large language models (LLMs), which are a specialised type of machine learning model that you can use to perform natural language processing (NLP) tasks, including:

- Determining sentiment or otherwise classifying natural language text.
- Summarizing text.
- Comparing multiple text sources for semantic similarity.
- Generating new natural language.

While the mathematical principles behind these LLMs can be complex, a basic understanding of the architecture used to implement them can help you gain a conceptual understanding of how they work.

Transformer models

- Transformer architecture extends successful techniques for modeling vocabularies in NLP tasks
- Training of transformer models with large volumes of text to represent semantic relationships between words
- Capability of transformer models to generate language responses indistinguishable from human responses
- Components of transformer model architecture: encoder block and decoder block
 - Encoder block: creates semantic representations of the training vocabulary
 - Decoder block: generates new language sequences
- Variations in transformer model implementations: BERT model uses only the encoder block, GPT model uses only the decoder block
- Nice illustration: <https://prvnsmph.github.io/animated-transformer/>

Tokenization

The first step in training a transformer model is to decompose the training text into tokens - in other words, identify each unique text value.

You can think of each distinct word in the training text as a token (though in reality, tokens can be generated for partial words, or combinations of words and punctuation)

Tokens will be used as the pricing units (will be discussed next week)

I (1)
 heard (2)
 a (3)
 dog (4)
 bark (5)
 loudly (6)
 at (7)
 ("a" is already tokenized as 3)
 cat (8)

		Unique ID
All words	"aardvark"	14
	"apple"	177
	"box"	392
	"cardboard"	477
	⋮	⋮

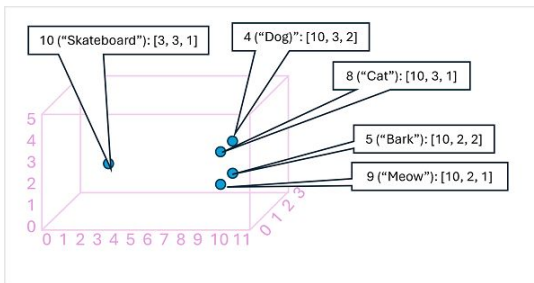
Transformer([3206, 2736, 3657, 400] , θ) $\rightarrow$ 2532

Embeddings

- Embeddings are contextual vectors assigned to tokens in natural language processing.
- They encapsulate semantic relationships between words by representing specific semantic attributes of each token.
- Embedding vectors position tokens in multidimensional space, with closer tokens indicating greater semantic relatedness.

Token embeddings

Source: <https://prvnsmph.github.io/animated-transformer/>



Attention

1. A sequence of token embeddings is fed into the attention layer. Each token is represented as a vector of numeric values.
2. The goal in a decoder is to predict the next token in the sequence, which will also be a vector that aligns to an embedding in the model's vocabulary.
3. The attention layer evaluates the sequence so far and assigns weights to each token to represent their relative influence on the next token.
4. The weights can be used to compute a new vector for the next token with an attention score. Multi-head attention uses different elements in the embeddings to calculate multiple alternative tokens.
5. A fully connected neural network uses the scores in the calculated vectors to predict the most probable token from the entire vocabulary.
6. The predicted output is appended to the sequence so far, which is used as the input for the next iteration.

Attention Is All You Need

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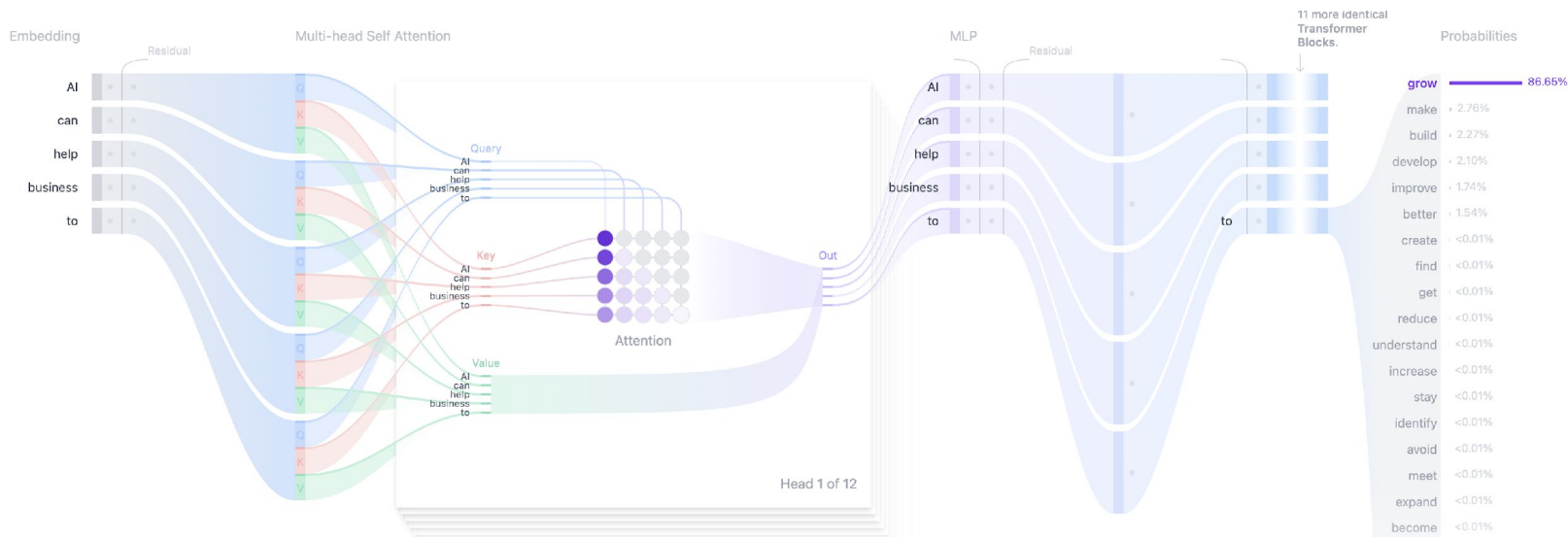
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Check out this amazing Transformer Explainer



Source: <https://poloclub.github.io/transformer-explainer/>

What are prompts (for GenAI)?



A starting point for
a conversation.



A specific request
or command.



A creative brief for
the AI.

How to prompt effectively? (1)

Be Specific & Clear:

- **Vague:** "Write a story."
- **Better:** "Write a short science fiction story (around 500 words) about a lonely robot on Mars who discovers an ancient alien artifact."
- *Clearly define your task, desired output, and any constraints.*

Provide Context:

- Give the AI relevant background information.
- **Example:** "I'm a marketing manager preparing a campaign for a new eco-friendly coffee brand. Generate three taglines that highlight sustainability and a premium taste."
- *Context helps the AI understand the 'why' behind your request.*

How to prompt effectively? (2)

Define a Role/Persona (for the AI):

- Instruct the AI to adopt a specific persona.
- **Example:** "Act as an experienced travel guide. Describe the top 3 must-see attractions in Paris for a first-time visitor interested in art and history."
- *This helps tailor the tone, style, and knowledge base of the response.*

Specify the Output Format & Structure:

- Tell the AI how you want the information presented.
- **Examples:**
 - "Provide the answer as a bulleted list."
 - "Write a formal email."
 - "Generate Python code for..."
 - "Summarize this article in three key paragraphs."
 - "Limit the response to 200 words."

How to prompt effectively? (3)

Use Examples (Few-Shot Prompting):

- Provide one or more examples of the desired input/output style or format.
- **Example:**
 - "Translate the following English phrases to French, in a formal tone:
 - English: Hello, how are you? -> French: Bonjour, comment allez-vous?
 - English: Thank you very much. -> French: Merci beaucoup.
 - English: Can you help me? -> French: ?"
- *This is very effective for guiding style and structure.*

Iterate and Refine:

- Prompting is often an iterative process. Don't expect perfection on the first try.
- Analyse the AI's output, identify areas for improvement, and refine your prompt.
- Try rephrasing, adding more detail, or simplifying.

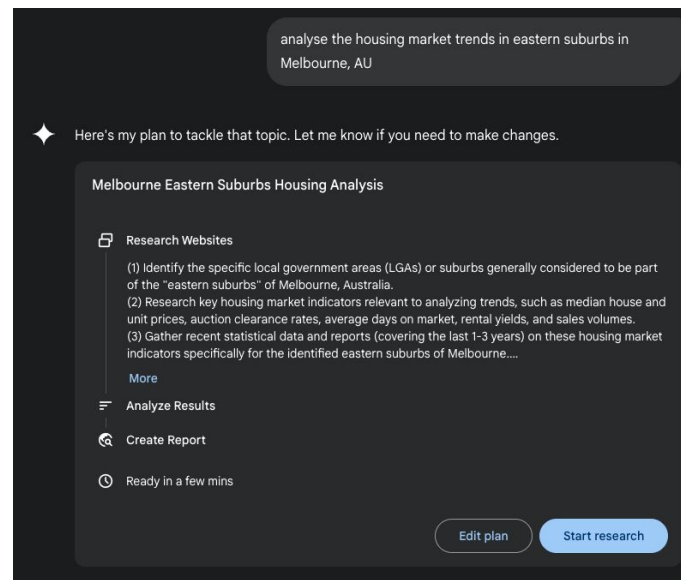
How to prompt effectively? (4)

Break Down Complex Tasks:

- For complicated requests, use a series of prompts or guide the AI step-by-step (Chain-of-Thought prompting).
- **Example:**
 1. "Outline the key arguments for renewable energy."
 2. "Now, for each argument, provide a supporting statistic."
 3. "Finally, write a concluding paragraph summarizing the importance of transitioning to renewable energy."

Use AI to do research?

- Some tools can help you run quick desk-research (understand markets, understand your potential client, understand a product, etc.)
- Tools with Deep Research capability: Gemini, ChatGPT, Perplexity



conduct a research for the following key points
for Global-E: (1) Global-e's key differentiators
(MOR, tax, APMs, payments, fraud) (2)
Objection handling (Markets, cost, control) (2)
Key personas (CFO, Head of eComm,
Marketing, Legal) (4) AU case studies: St. Agni,
Muscle Republic, Ryderwear.



Here's my plan to tackle that topic. Let me know if you need to make changes.

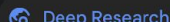
Global-E Research Key Points



Research Websites

- (1) Obtain a comprehensive overview of Global-E's business model and the suite of services it offers for cross-border e-commerce.
- (2) Analyze Global-E's core differentiators, specifically examining its capabilities as a Merchant of Record (MOR), its approach to international tax and duties, its support for various Alternative Payment Methods (APMs), its overall payment processing solutions, and its fraud prevention strategies.
- (3) Investigate how Global-E addresses common client objections concerning market reach and expansion, the cost

Enter a prompt for Gemini



Global-E Research Key Points

Show thinking



Research Approach

I'm starting a comprehensive research project on Global-E, aiming to cover its key differentiators, strategies for handling objections, target personas, and specific Australian case studies. I'll begin by gaining a general understanding of Global-E's business model and services, and then progressively delve into each specific area of inquiry.

Initial Focus

My initial focus will be on understanding Global-E's core differentiators, such as its merchant of record (MOR) model, tax calculation and remittance processing, and fraud prevention.



Researching websites



investor... Investor Relatio...



global... Global E-Commer...



global... Drive your e-com...



Understanding Global-E's Core Offerings

I've started by gathering information about Global-E's business model and the suite of services it offers for cross-border e-commerce. I've identified that simplifies and accelerates global expansion, providing a localized shopping experience in over 190 countries to handle all aspects of international trade.

Initial Insights on Differentiators

My initial searches have provided some insights into Global-E's Merchant of Record (MOR) model, tax calculation and remittance processing, and fraud prevention. These services, while beneficial, can have complexities, and Global-E emphasizes the importance of understanding these complexities.

Next Steps in Research

To deepen my understanding, I'll now focus on their differentiators. Concurrently, I'll investigate their target markets, cost, and control, and how they address common client objections. Finally, I'll search for specific Australian case studies involving St. Agni, Muscle Republic, and Ryderwear.

Global-E: A Strategic Enabler for Cross-Border E-commerce Dominance

Executive Summary

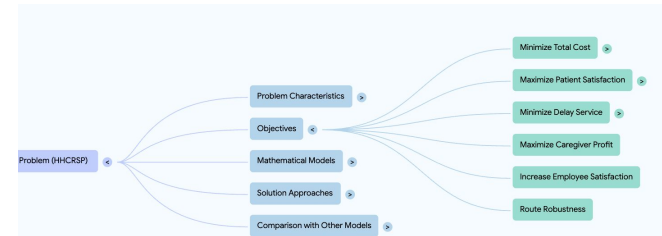
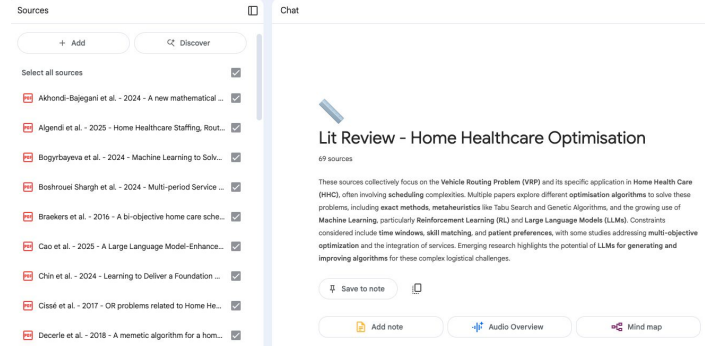
Global-E stands as a leading enabler and accelerator of global, direct-to-consumer (DTC) cross-border e-commerce growth. Its comprehensive platform simplifies international selling by merging three critical pillars: Global Enablement, Global Intelligence, and Global Demand, allowing merchants to enhance the buying experience, streamline operations, and significantly boost e-commerce revenues worldwide.¹ This report will demonstrate how Global-E's unique differentiators, particularly its robust Merchant of Record (MOR) model, sophisticated tax and payment solutions, and advanced fraud prevention, directly address common merchant concerns regarding market reach, cost, and control. Furthermore, it will detail tailored value propositions for key organizational personas and showcase tangible success through Australian case studies.

1. Global-E's Foundational Platform and Business Model

Global-E's business model is fundamentally about "Powering better global e-commerce, transforming how merchants and shoppers across the world engage directly".¹ The company partners with leading merchants globally to enhance the buying experience, simplify global expansion, and boost e-commerce revenues, enabling brands to unlock worldwide potential and own customer relationships globally.¹ This strategic partnership approach allows businesses to navigate the complexities of international trade with greater ease and effectiveness.

NotebookLM

- NotebookLM is a useful tool to extract information from different sources
- Can work with website, pdf files, google docs, YouTube videos
- Responses will be grounded with the sources
- Good ways to pull things together





What are you interested in?

Help me understand Australian Footwear brands that can potentially go international

Australian Footwear Brands: Global Exp & Marketing

10 sources

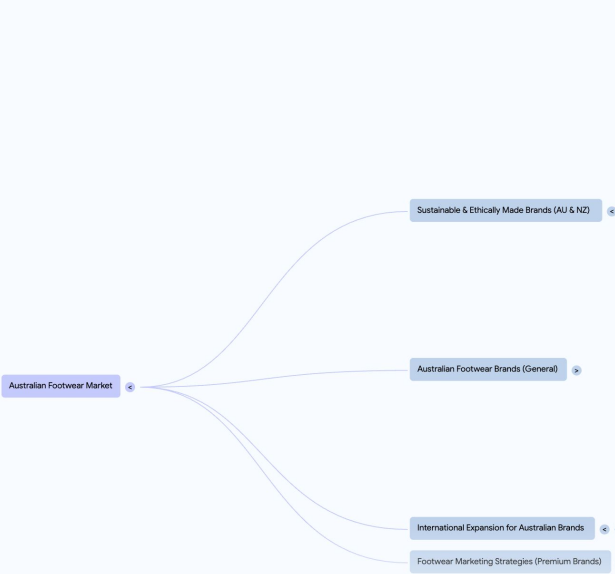
These sources collectively discuss the **Australian footwear market**, highlighting its **growth** and through various brands like Ugg. One article **showcases specific Australian shoe labels**, noting designs, celebrity appeal, and commitment to quality or sustainability. Another source provides for premium footwear brands, emphasising the importance of **digital presence**, **brand story** for international success in a booming market. Finally, two sources address the broader trend of **businesses expanding internationally**, detailing the **challenges and strategies** involved in moving from domestic market, from **market fit** and **regulatory hurdles** to **financial planning** and **operation** particular focus on the footwear industry's potential for global reach.

Save to note

Add note

Audio Overview

Bad content



These sources examine Australian footwear brands with international potential, covering market expansion, challenges, brand identity, and global strategies.

Select all sources

Two in three Australian brands are looking abroad to scale up. Here's why - Inside Retail

Explains why Australian brands, including footwear, are expanding globally.

13 Of The Best Australian Shoe Brands Worth Investing In - Grazia

Introduces top Australian shoe brands with existing international recognition.

A deep dive into Ugg, Australia's most influential streetwear export - Fashion Journal

Details the global strategies success story of Australian Ugg boots.

Discuss what these sources say about Bared Footwear, in the larger context of Australian Footwear Brands (General).

Bared Footwear is highlighted as a distinct and noteworthy Australian shoe brand, particularly due to its unique foundation and design philosophy 1.

In the larger context of Australian footwear brands (General), the sources reveal several key characteristics of the market:

- **Growth and Influence:** Despite Australia's smaller population compared to global style hubs, the Australian footwear market is described as "surprisingly dense" and consistently growing, with new labels emerging regularly 2 3. Australian shoe brands are gaining international recognition, being seen on celebrities and red carpets globally 3.
- **Focus on Quality and Comfort:** Australian brands are in the business of producing not only stand-out styles from a fashion perspective but also **comfortable, well-made footwear that is designed to last beyond a single season** 3. This emphasis on longevity is a recurring theme among many Australian brands 3 ...
- **Sustainability and Ethical Practices:** There is a strong, impactful trend in footwear marketing towards sustainability and ethical practices, driven by increasingly conscious consumers 6. Brands that highlight eco-friendly materials, fair trade practices, and supply chain transparency can build trust and loyalty 6. While Teva (US-based) is specifically noted for its sustainability initiatives like using recycled plastic 7, several Australian brands also

Cloud services for GenAI

- Generative AI and Cloud Platforms: Generative AI requires massive computing power and large datasets, making the public cloud an ideal platform.
- Key Players: The top hyperscalers — Amazon Web Services, Google, and Microsoft — are in intense competition in the GenAI space.
- Amazon Web Services (AWS): AWS is investing in three key services for GenAI — Amazon Sagemaker JumpStart, Amazon Bedrock, and Amazon Titan.
- Google Cloud: Google Cloud also offers services for Generative AI.
- Infrastructure Providers: Infrastructure providers offer scalable, reliable, and cost-effective solutions for generative AI projects.

Case Study: Amazon Review

- Let's consider a scenario where a business aims to **analyse the customer reviews** for their electronics products sold on Amazon.
- Assume this business wants to process approximately **10,000 customer reviews each day**. Based on the earlier finding that the average length of an electronics product review on Amazon is around **209 characters** and using a conservative estimate of approximately **4 characters per token**, each review would roughly translate to **53 tokens**.
- For each review, we might want to extract a **sentiment label** (which would typically be a single token) and **identify up to three key topics**. If each topic is represented by a short phrase averaging around 5 words, this would translate to approximately **7 tokens per topic**. Thus, the output for topics would be 3 topics multiplied by 7 tokens per topic, totalling 21 tokens. Adding the 1 token for sentiment, the **output per review would be approximately 22 tokens**.

Cost Analysis

Feature	OpenAI (GPT-4o)	OpenAI (GPT-3.5 Turbo)	Google (Gemini 1.5 Flash)	Google (Gemini 1.0 Pro)
Input Price (per 1M tokens)	\$2.50	\$0.50	\$0.075	\$0.50
Output Price (per 1M tokens)	\$10.00	\$1.50	\$0.30	\$1.50
Monthly Input Cost	?	?	?	?
Monthly Output Cost	?	?	?	?
Total Monthly Cost	?	?	?	?

Assume 30 days per month?

Which LLM model can minimise our cost?

Activity: How to control the responses of LLMs

- Most available GenAI tools are designed as a chatbot with general responses and try to adapt as the conversation progresses
- As the responses from AI are not deterministic, you can get different outputs from different interactions with AI even through the prompts are the same
- For casual use, it is fine but you might not want arbitrary responses for serious work (e.g. analyse sentiment for a customer review)
- For a task, if you have a workflow and certain expected output formats (images, python code, diagrams, tables etc.), you should pass them to the AI

Use AI to automate your workflows

- Describe your workflow to AI
- Be clear about the inputs provided to AI and your expected outputs
- Simple Example:

This document will provide the instructions that you must follow.

You will need to follow the steps below to collect data from a person.

Step 1: Ask full name

Step 2: Ask occupation

Step 3: Summarise provided details

Step 4: Back to step 1 to collect data from another person

When user say "All done", stop asking and present the table with all details collected.

 **New Gem** Save

Name

Preview


To preview your Gem start by giving it a name

+

Instructions ⓘ

Example: You are a horticulturist with a background in natural lawns and native plants, and you help people plan low water gardens. Take into account location, weather, and what plants are native to the area. You are knowledgeable, casual, and friendly.

Knowledge ⓘ

+

ⓘ Gemini can make mistakes, so double-check responses. Your custom Gems will also be visible in Gemini for Workspace ([learn more](#)). Create Gems responsibly.

Test your system prompt with Gemini (or Val)

Practice: Personalised Chatbot with Val

- Reimplement the chatbot above with Val
- Login RMIT Val
- Go to My Studio
- Click Create Persona
- Enter your the prompt to system prompt
- Click the newly created persona to test



Reset Image

Testing System Prompt

testing-system-prompt_5sa0r2

Default Base Model
OpenAI GPT-5.2

Description
Test how the structured outputs are designed

+ Add Tags

Share Links
No share links created yet

Persona Params

System Prompt
This document will provide the instructions that you must follow. You will need to follow the steps full name Step 2: Ask occupation Step 3: Summarise provided details Step 4: Back to step 1 to co done", stop asking and present the table with all details collected.

Advanced Params

Prompt suggestions

Capabilities
☒ Vision

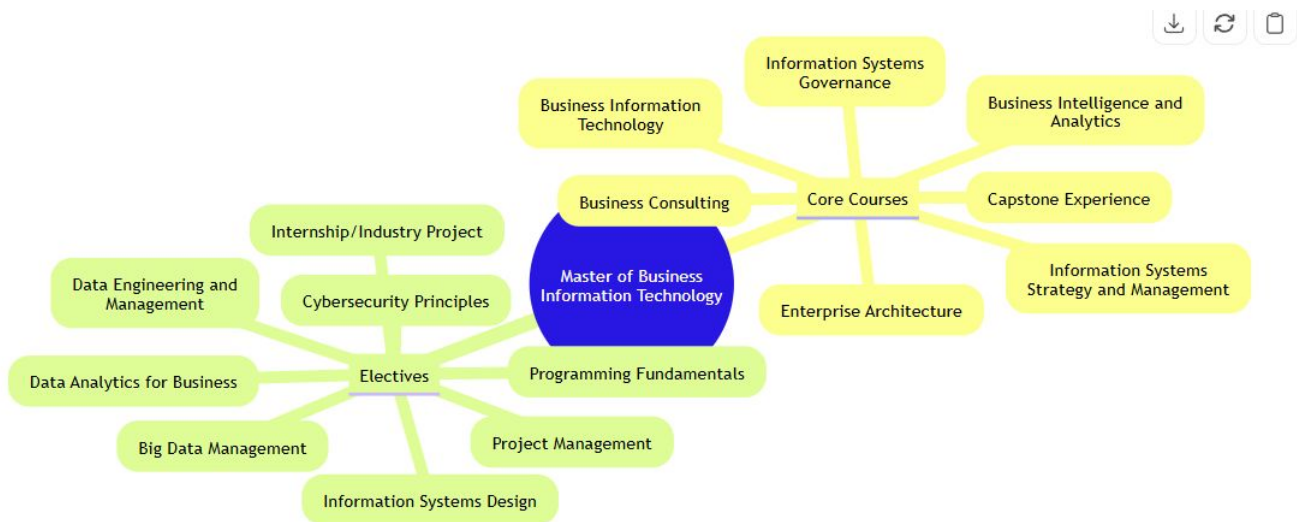
Save & Update

Group Activity: News Analytics

- Work as a group (your table), and identify a hot topic that the news has covered recently. For example, **“Home prices rise as first-time buyers, investors see 'intense competition'”**
- Build a Chatbot to help you analyse the news (can include both texts and images), and transform them into a desired format (table with summary, key authors/publishers, date)
 - Define your workflow
 - Think of how you pass information/data to chatbot (attached files, copy texts, etc.)
 - Define the output format
 - Design your system prompt
- Present your chatbots to the class

Val can display output as infographics

“create a mindmap of program structure (including course names) for master of business information technology at RMIT with Mermaid.”

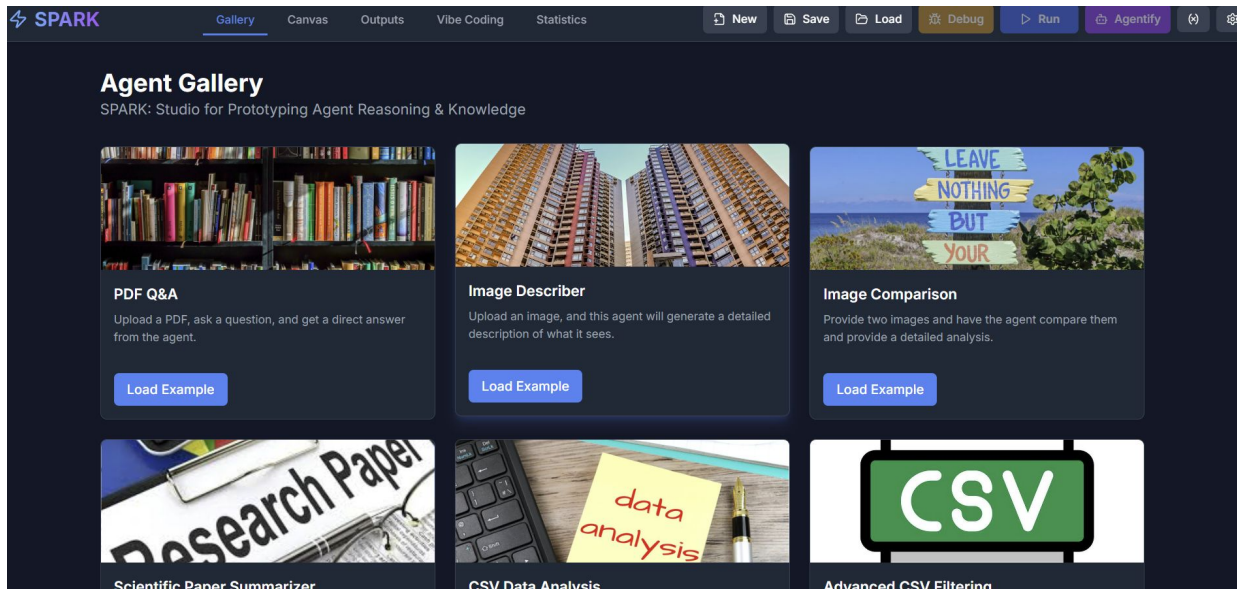


What do you think of the outputs?

How to improve?

Activity: Explore SPARK tool

- SPARK stands for “Studio for Prototyping Agent Reasoning and Knowledge”
- Access via: <https://staging.d2z74xh3q5fafu.amplifyapp.com/>



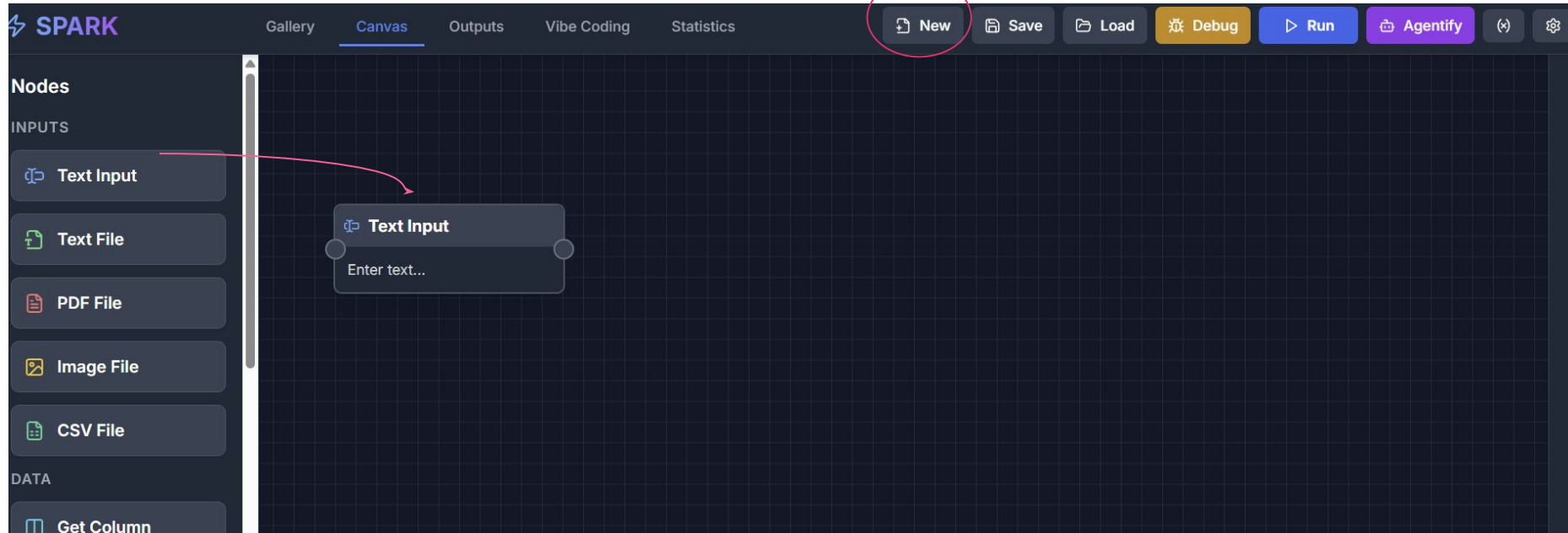
It is still under development so there might be errors or technical glitch.

How is SPARK different from Val, ChatGPT, etc.

- SPARK is model-agnostic (designed to work with different LLMs, including local models)
- SPARK is built for training/education purposes
- SPARK allows user to easily customise how they want to use LLMs, not just as chatbots
- SPARK comes with a number of tools that allows users to explore the capabilities of LLMs
- SPARK allows users to design single or multiple workflows in a workspace
- SPARK can build AI agents based on existing workflows

Get Started

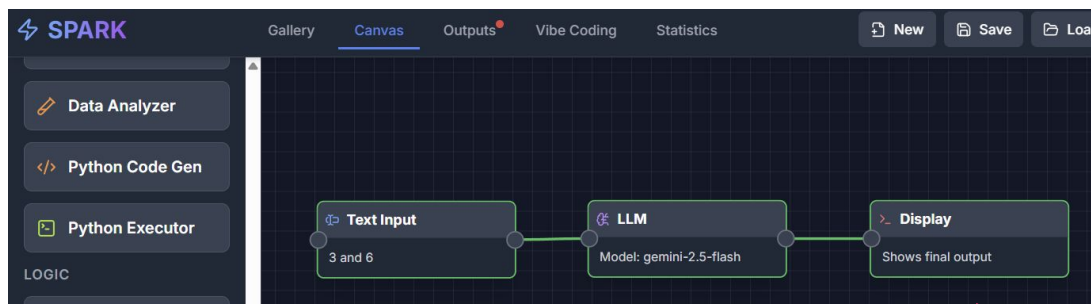
Start here



Build a flow by dragging nodes from the left pane to the canvas

Join the output of one node to the input of another node to create connection

Build a simple flow



Click the node

Click



PropertiesVariables

Node ID

node-0

Name

Text Input

Input Text

3 and 6

LAST RUN DATA

View Input

View Output

Delete Node

PropertiesVariables

Node ID

node-1

Name

LLM

AI Provider

Google Gemini

Model / Deployment Name

gemini-2.5-flash

Output Format

Text

Max Output Tokens

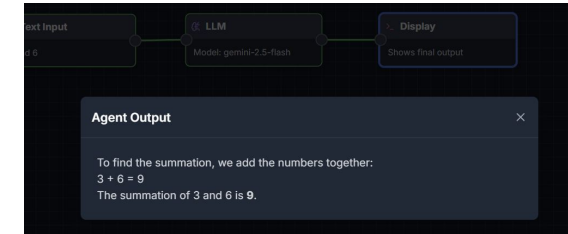
2048

Prompt

Use variables like {node-id} to reference inputs.

Provide the summation of the numbers below:

{node-0}



- Make sure to provide Gemini API key.
- Review this [page](#) if you have not yet got your key (Google Account is required)

Settings

AI Provider

Google Gemini

Google AI API Key

AlzaSyAikAKBU2m6MnPIVqm3LTWpi6R5VCDn4Js

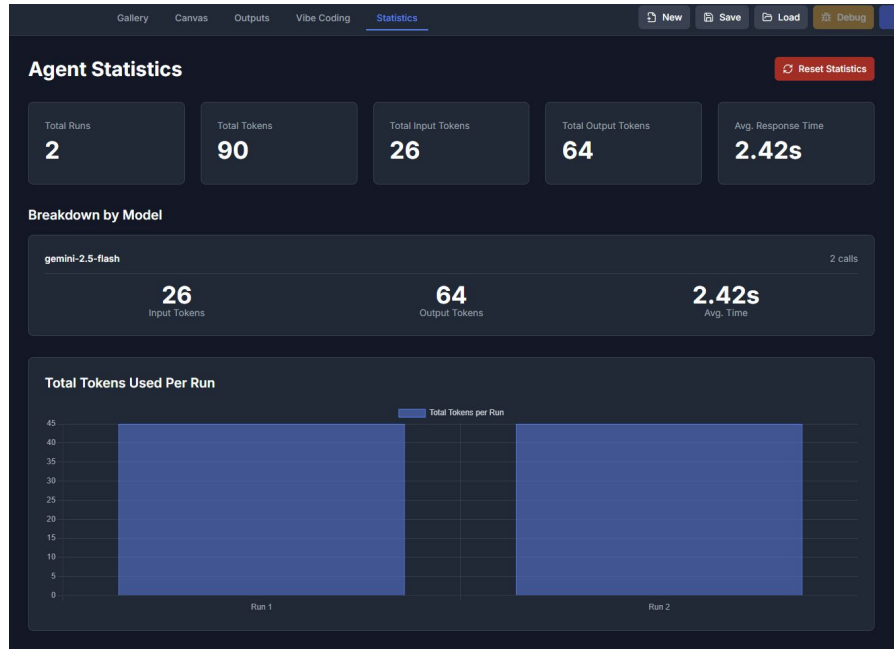
Pyodide Packages

pandas,matplotlib,scikit-learn,numpy

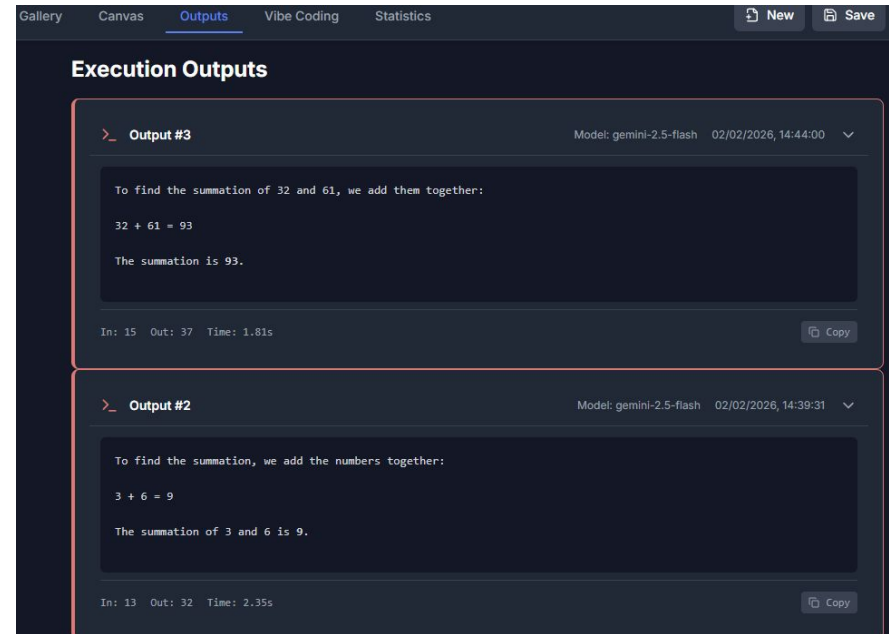
Comma-separated list of Python packages to load at startup (e.g., pandas,numpy,matplotlib).

Save Settings

See statistics



Review Outputs

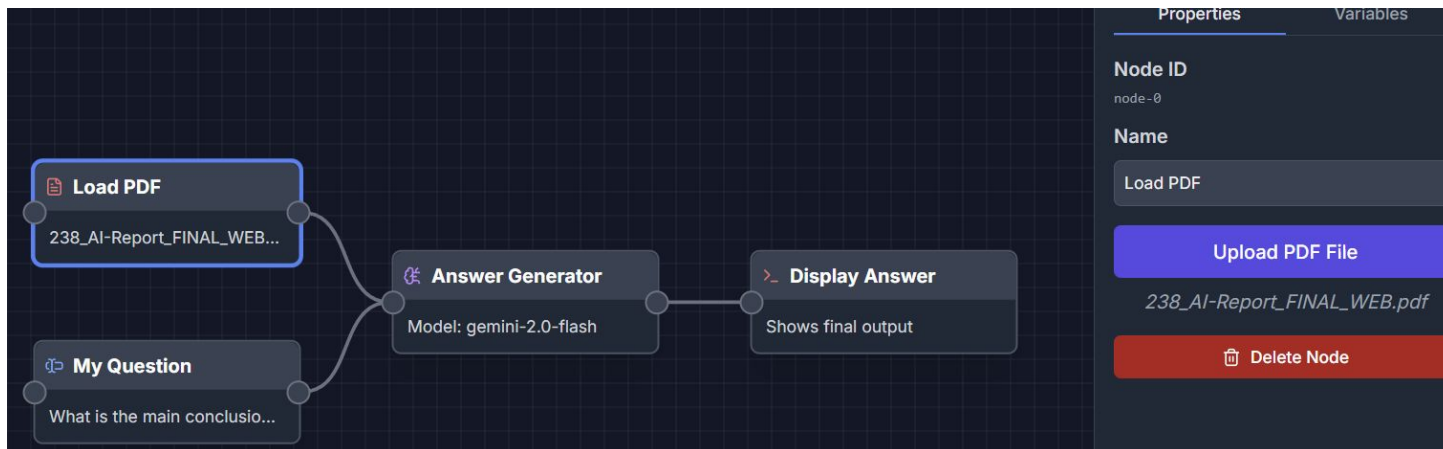


Increase the complexity

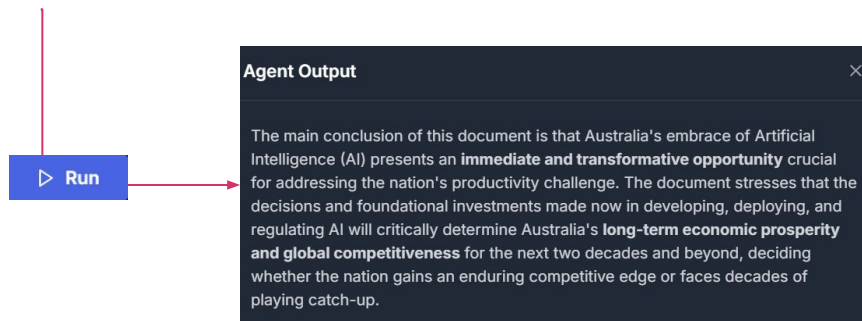


Handling Files with SPARK

Load the first
example in Gallery



Upload the "Accelerating Australia's AI Agenda" report in the Load PDF node.



Put all together

- Modify the SPARK flow in the first example to present the key action points in a user and concise format.
 - Quickly go through the report
 - Design the profile of the audience you would like to present to
 - Design the output format
 - Modify the prompts in the flows
 - Run and discuss the results

Assessment 2 Overview - Context

Modern Furniture Co. is a custom furniture manufacturer looking to leverage its operational data to become more profitable and efficient. The company uses a sophisticated business simulator that generates rich data across all facets of its operations, from sales and finance to production and human resources. However, the management team lacks the tools to easily query this data to get timely, actionable insights.

Your task is to propose and build a proof-of-concept for an AI Analytics Agent. This agent will act as an intelligent assistant, allowing a manager to ask questions in plain English like:

- "What was our biggest production bottleneck last quarter?"
- "Show me the impact of employee resignations on our on-time delivery rate."
- "Which product type is the most profitable?"

The agent should process these queries, analyse the relevant simulator data, and provide answers that may include text summaries, key metrics, and visualisations.

Assessment 2 Overview - Objectives

- To provide an interactive and intuitive way for non-technical business users to explore operational data.
- To deliver timely and relevant insights to address the core challenges identified in the case study (e.g., demand forecasting, production efficiency, cost control, reputation management).
- To leverage generative AI technology to understand user intent and generate insightful responses based on the provided datasets.
- To assist management in making strategic, data-driven decisions to improve the overall performance of the business.

Assessment 2 Overview - Data

- You will be provided with a dataset generated by the Advanced Furniture Manufacturing Simulator. This dataset includes multiple CSV files such as a Financial Ledger, All Orders, HR Events Log, Inventory Log, etc. Your AI agent must be designed to integrate with and analyse this data.

Assessment 2 Overview - Submissions

- Slide deck and short demonstration video

